

Tracking Internet Disruptions in Ukraine: Insights from Three Years of Active Full Block Scans

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 netidee
FÖRDERUNGEN



FWF Österreichischer
Wissenschaftsfonds

22-07

2022-08

2022-09

2022-10

2022-11

2022-12

2023-01

Motivation

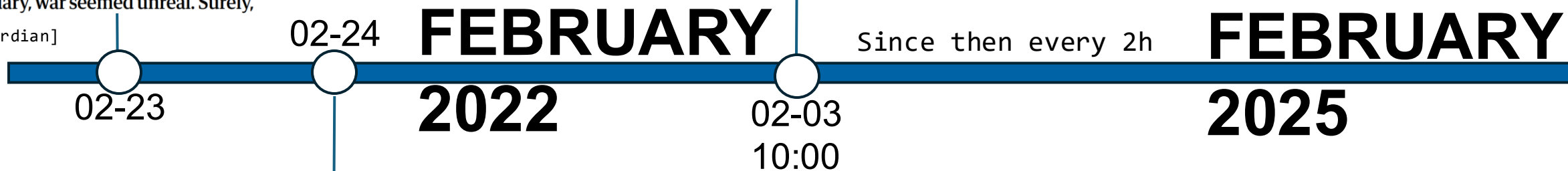
The night everything changed: waiting for Russia's invasion of Ukraine

Despite all the warning signs, as I sat down for dinner with friends in Kyiv on 23 February, war seemed unreal. Surely,

[Guardian]

Assumption: Internet-connected devices serve as sensors

```
$ countrymonitor/scan.sh UA 8000
```



The Invasion of Ukraine: How Russia Attacked and What Happens Next

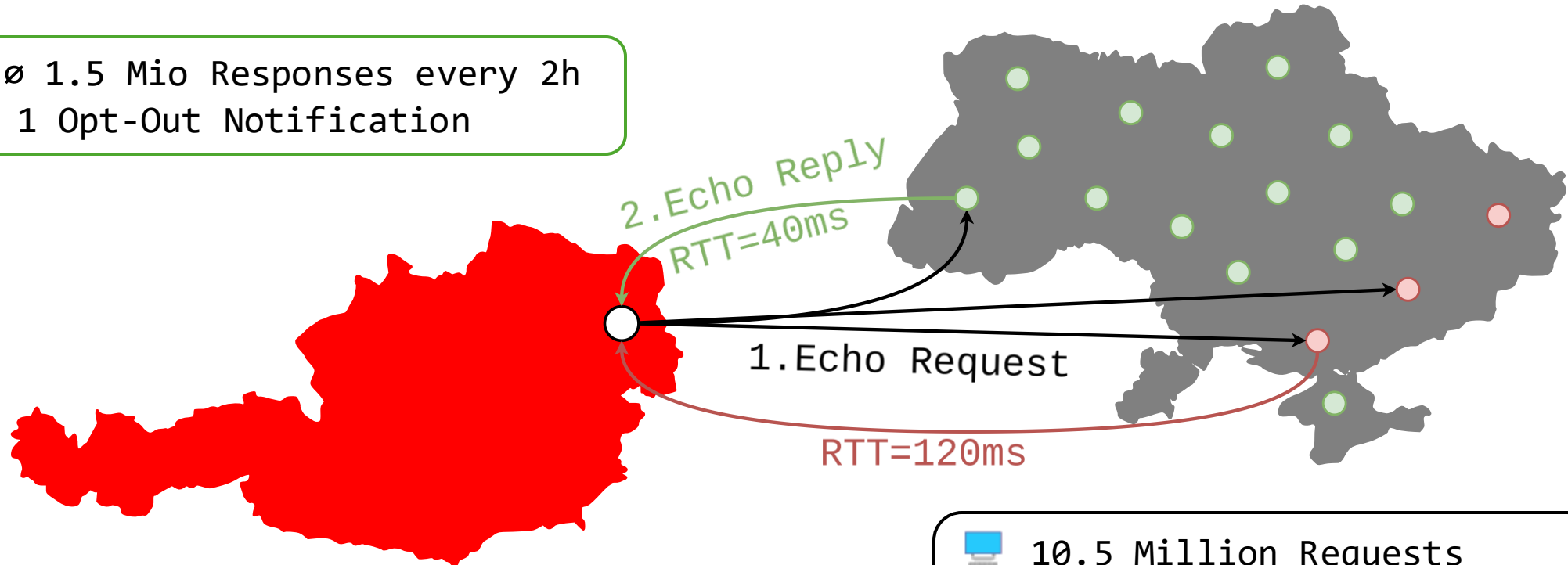
Explosions thundered in Ukrainian cities shortly after President Vladimir V. Putin declared the start of a “special military operation.”

[New York Times]

From Austria to Ukraine and back



∅ 1.5 Mio Responses every 2h
1 Opt-Out Notification



`aim.sba-research.org`



10.5 Million Requests



8,000 Packets per Second



22 Minutes Scan Duration

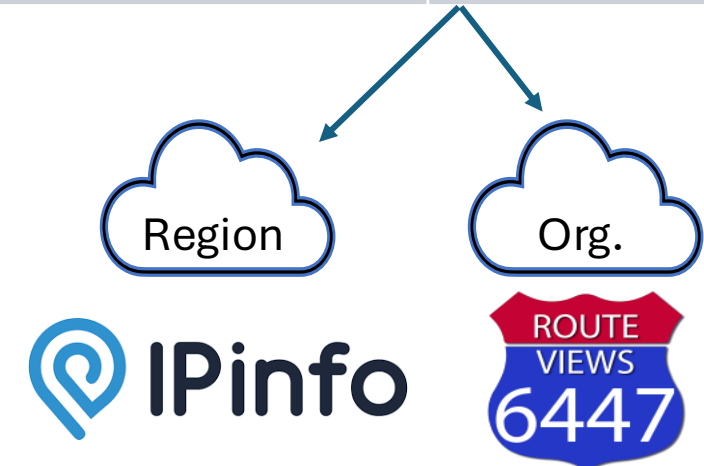
From Ping to Outage Signal

1. IPs to /24 blocks

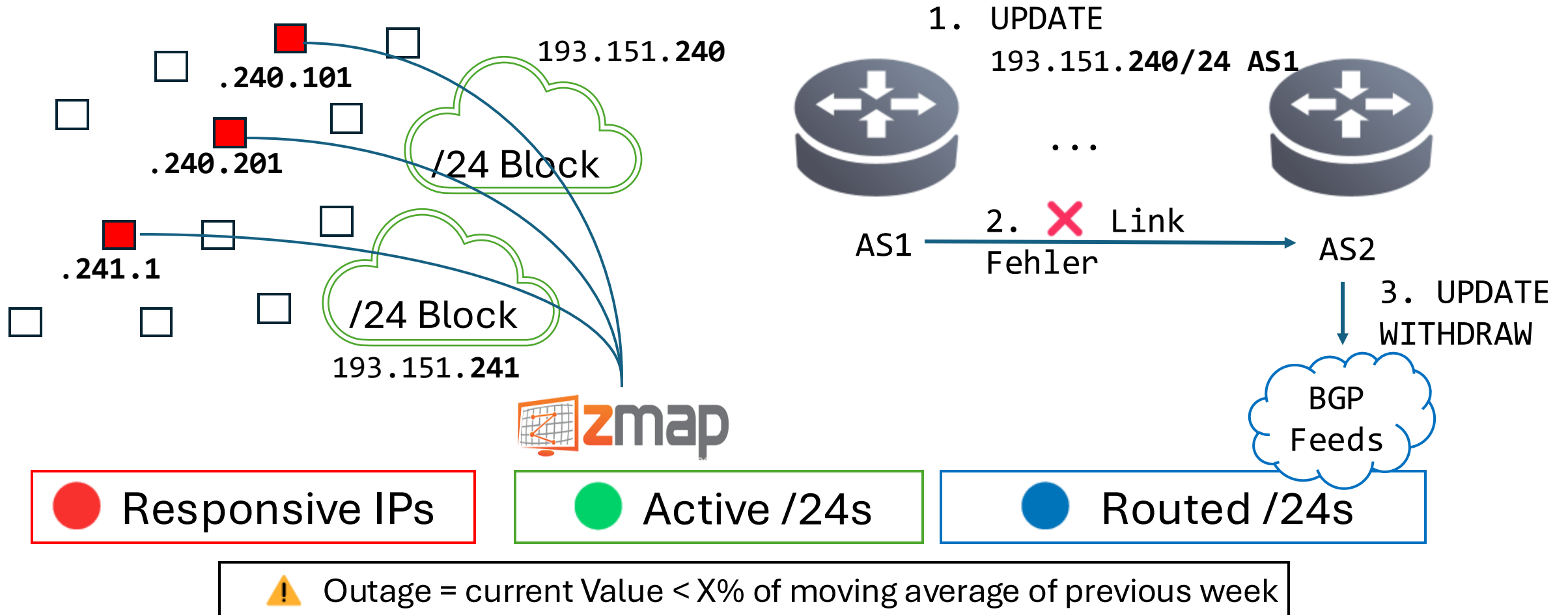
Timestamp	IP	/24
2022-03-02 10:00	193.151.240.1	193.151.240
2022-03-02 10:00	193.151.240.112	193.151.240
2022-03-02 10:00	193.151.240.13	193.151.240
2022-03-02 10:00	193.151.240.15	193.151.240
2022-03-02 14:00	193.151.240.1	193.151.240
2022-03-02 14:00	193.151.240.112	193.151.240
2022-03-02 14:00	193.151.240.15	193.151.240

2. Block attribution with external datasets

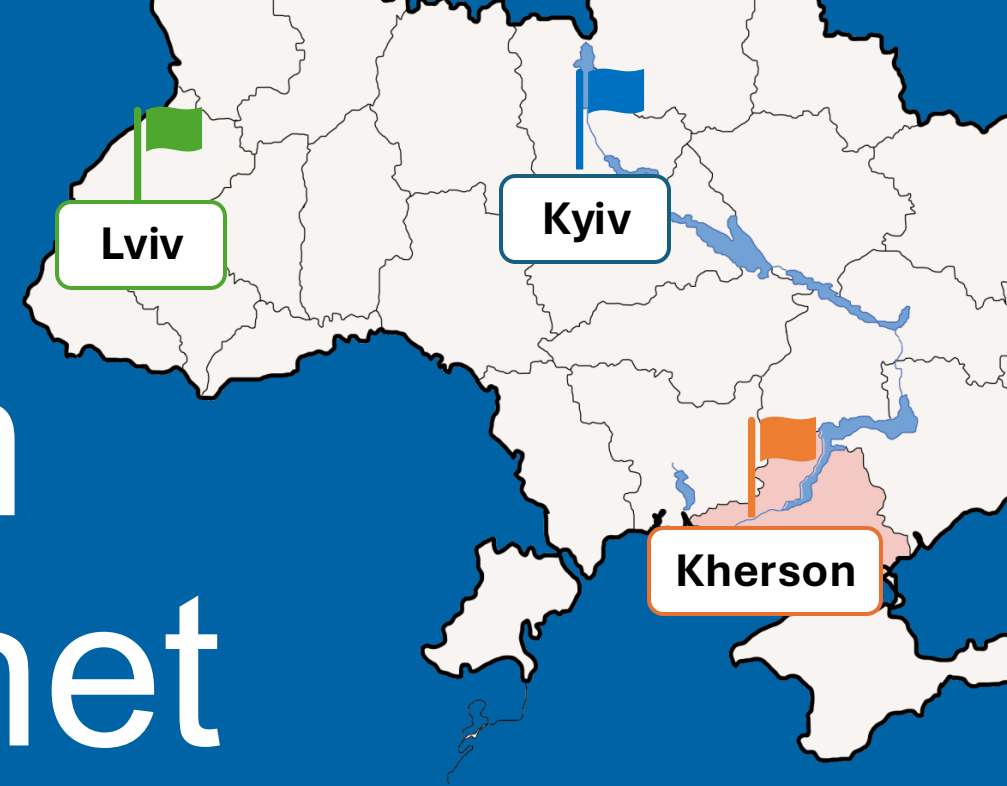
Timestamp	/24	
2022-03-02 10:00	193.151.240	✓
2022-03-02 12:00	193.151.240	✗
2022-03-02 14:00	193.151.240	✓



From Ping to Outage Signal



8 Insights from Tracking Internet Disruptions in Ukraine



#1 Existing measurements fail to comprehensively track disruptions in Ukraine.



	Singla et al.	Klick et al.	IODA
Protocols	Modbus, DNP3	60+	ICMP
Signal	IP based	IP based	Block based (Sampling)
Interval	24 Hours	4 Hours	10 Minutes
War Coverage	6 Months 2022	Until March 2023	Since 2022
Outage Location	IP Geolocation	Static IPs	IP Geolocation

Trinocular



Signal differences?
Short-lived outages?

Limited war coverage
Static IPs only?

Sampling?
Geolocation Precision?

#2 We report outages for many smaller ASes in Ukraine while IODA does not.

IODA returns raw data for **90%** of ASes (1,597 out of 1,773) but only reports outages for **19%** (333) of ASes.

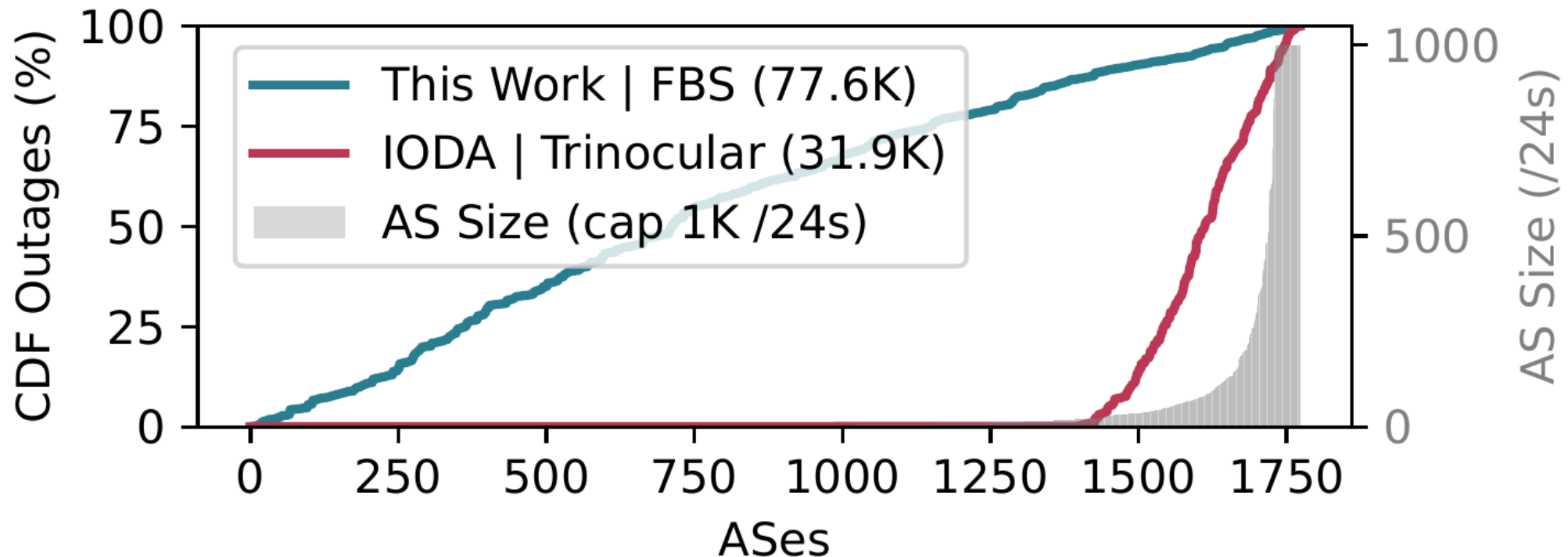


Figure 2: Outages reported for 1,773 ASes with regional /24s in Ukraine.

#3 Significant IP churn visible in Ukraine, especially in frontline regions.

60% between
oblasts

Ukrtelecom, Kyivstar,
Vodafone, Vega

40% to other
countries (1.5M IPs)

33% **amazon**

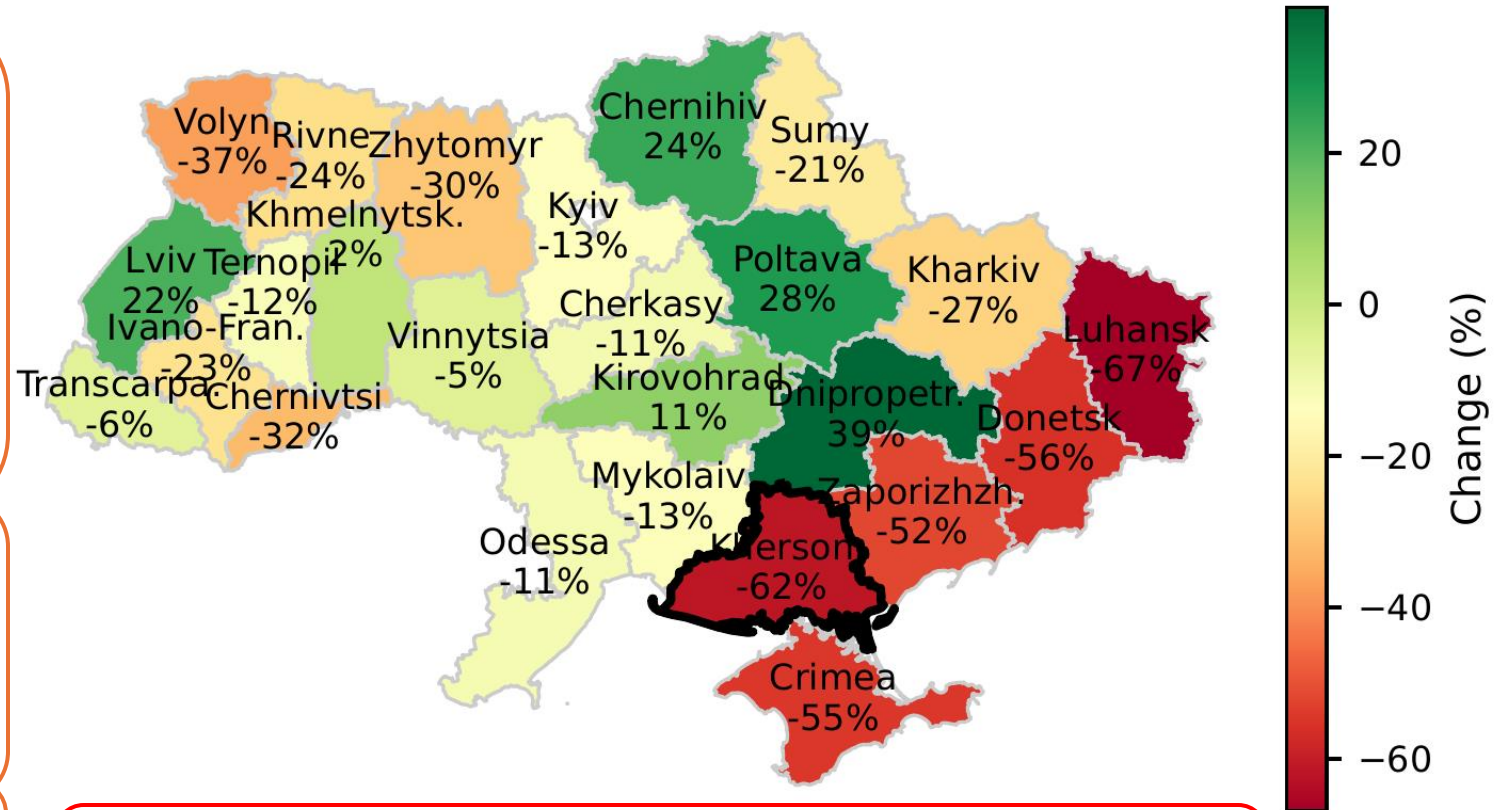


Figure 3: IP Churn - Comparing 2022 with 2025
3.7M IPs affected

#4 Regional classification detects long-term trends in geolocation.

Challenge:

Assign outages to a region in Ukraine



Regional Classification: Only consider /24s that have more than M addresses in the target region for more than T_perc of routed months.

In the paper: $M = T_perc = 70\%$

Timestamp		/24	
2022-03-02	10:00	193.151.240	✓
2022-03-02	12:00	193.151.240	✗
2022-03-02	14:00	193.151.240	✓

#4 Regional classification detects long-term trends in geolocation.

Class	Kherson		
	<i>ASNs</i>	<i>IPs</i>	<i>/24s</i>
Total	118	73.8K	512
Regional	13	8.3K	33
Non-Regional	40	65K	465
Target Set	34	41.7K	168

Kherson

1. Identifies 32% of blocks for which outages are representative of the Kherson region
2. Target set serves as input to outage detection.

#5 Internet disruptions in Ukraine peaked during winter 2022/23 and throughout 2024.

1. Intense attacks on the power grid -> partial outages through IPS signal
 - Infrastructure types (core vs edge)
2. Full block outages (FBS) more frequent in frontline regions
3. Only part of the outages are passively visible in BGP

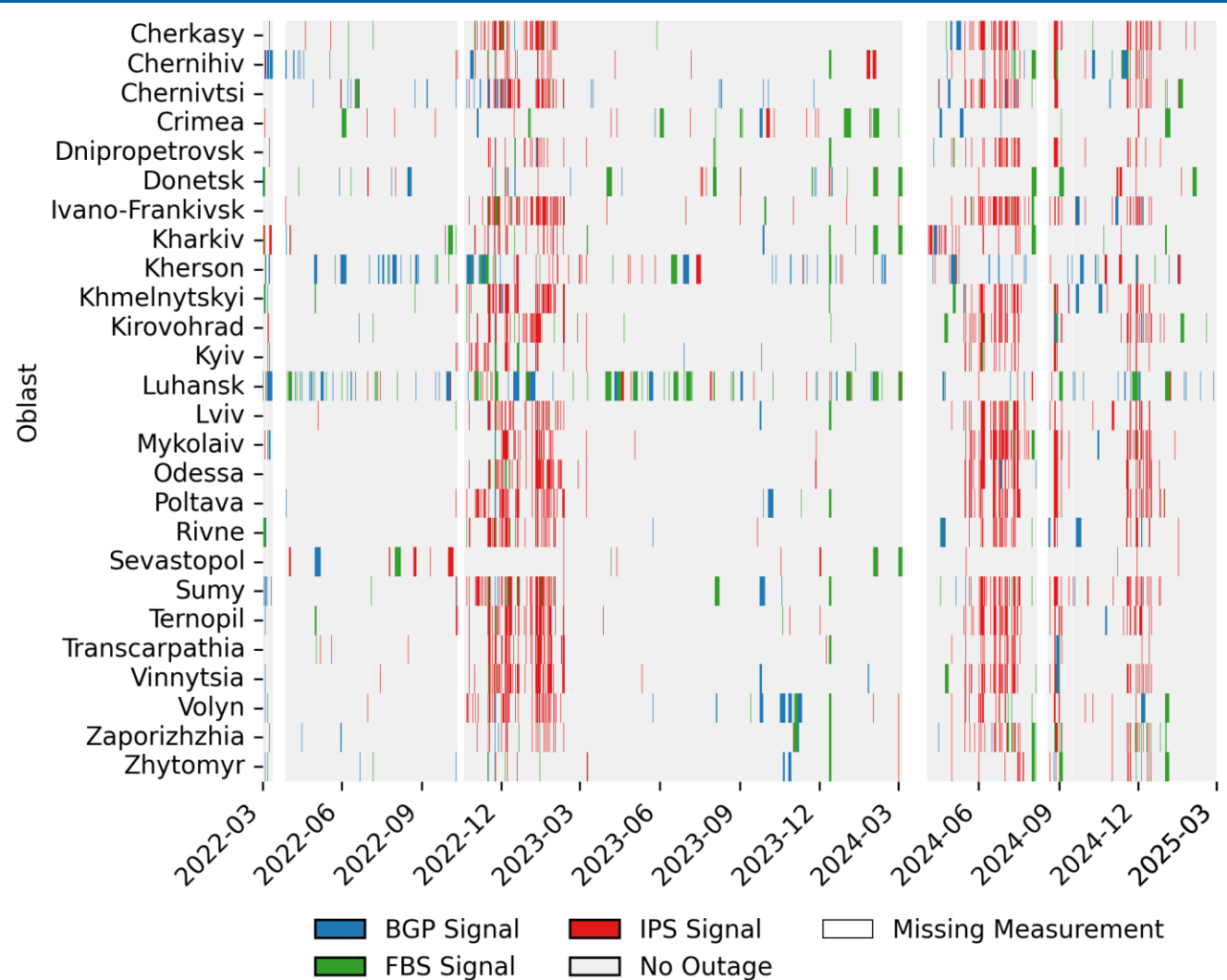


Figure 5: Outages detected by each signal type across oblasts.

#6 Difference between frontline and non-frontline unique to our dataset.

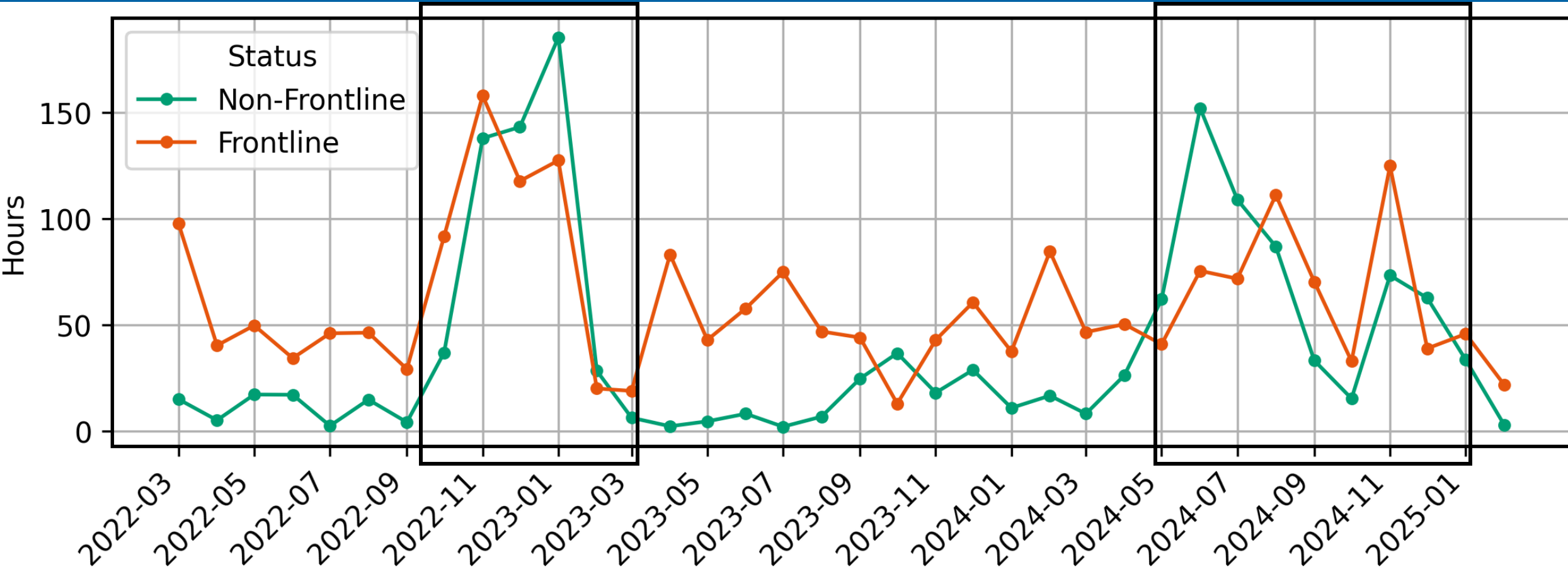


Figure 7: Average number of 2h intervals with outages.

#6 Difference between frontline and non-frontline unique to our dataset.

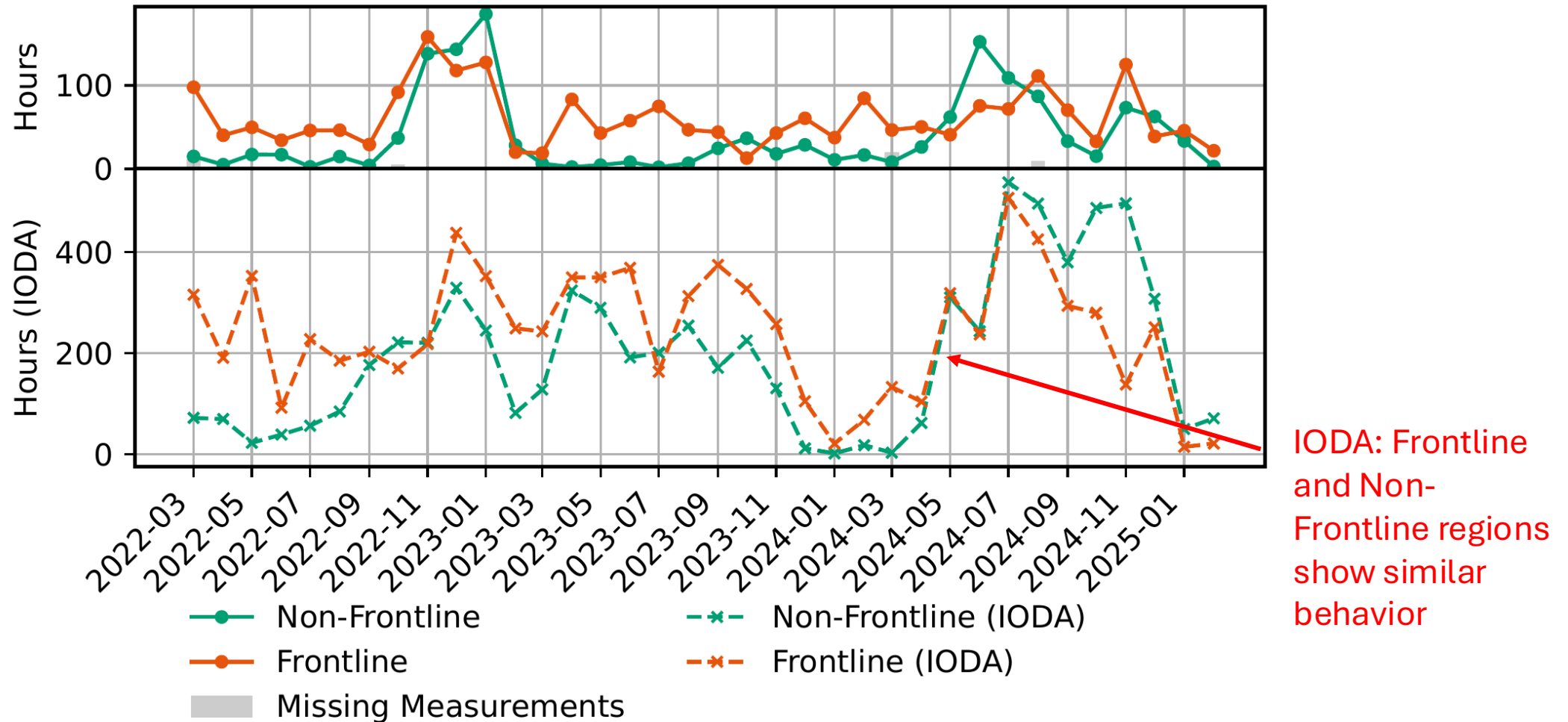


Figure 7: Average number of 2h intervals with outages.

#7 Strong correlation of days with outages & power cuts in non-frontline regions.

1951 HOURS
with electricity outages in 2024

Avg. 686 HOURS
Max. 2822 HOURS
with Internet outages in 2024

Pearson $r=0.725$
overlap of days

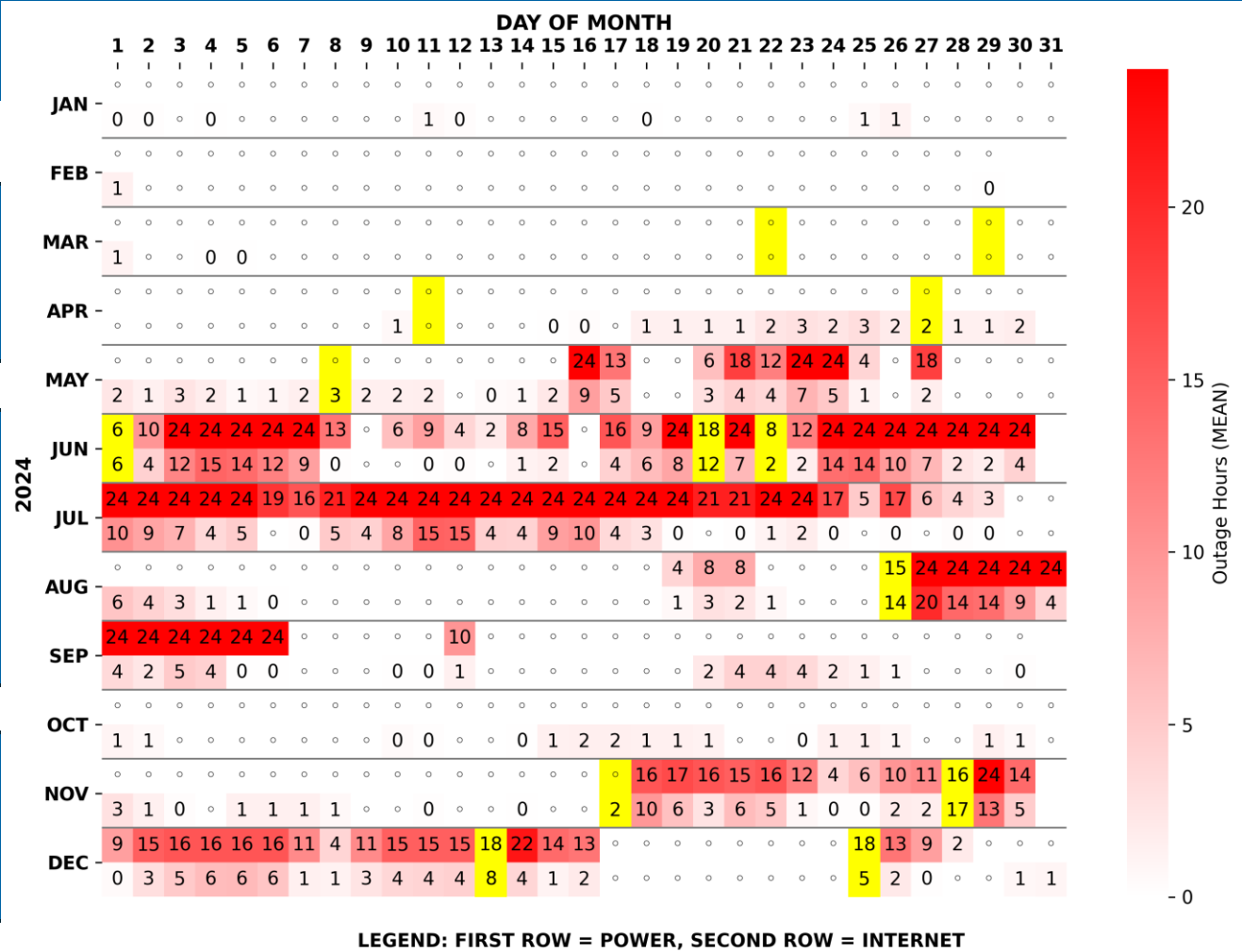


Figure 8: Hours affected by power and Internet outages in 2024.

Kherson – 3 Years on the Frontline

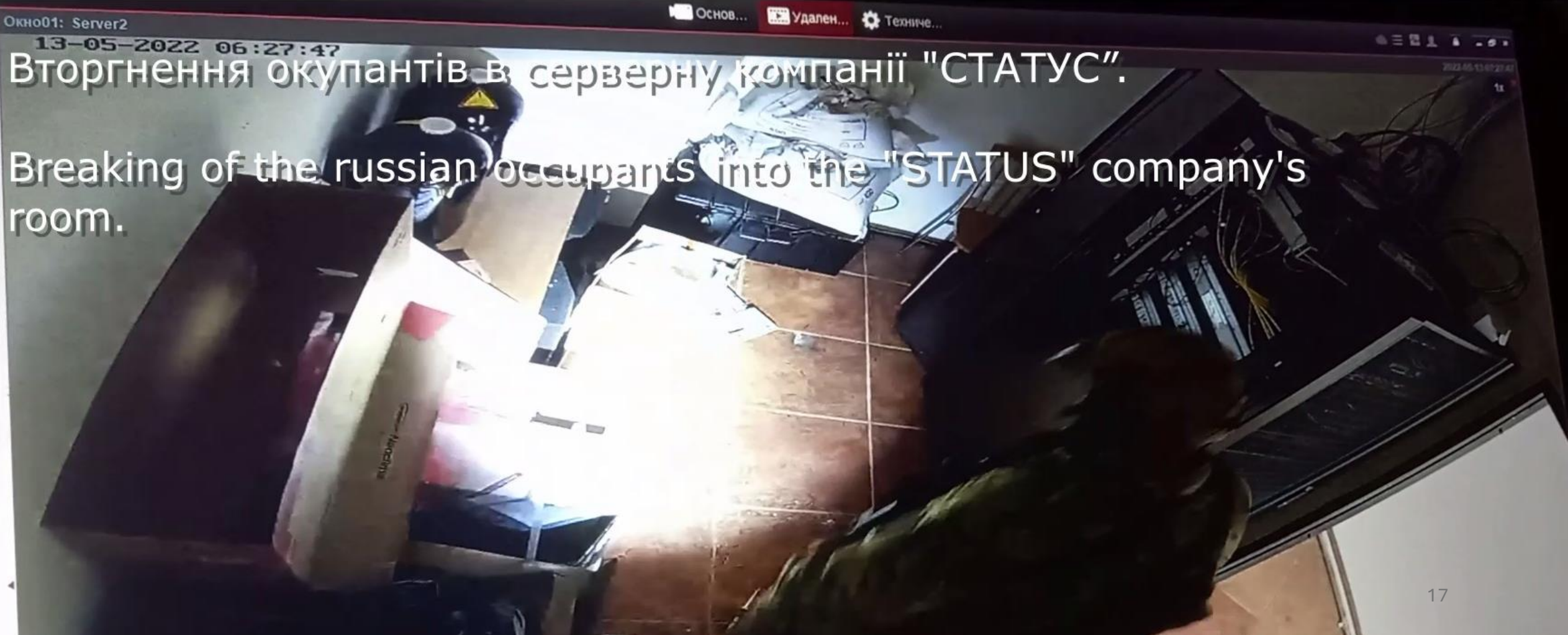


#8 Dataset provides insights into smaller regional providers in Kherson.

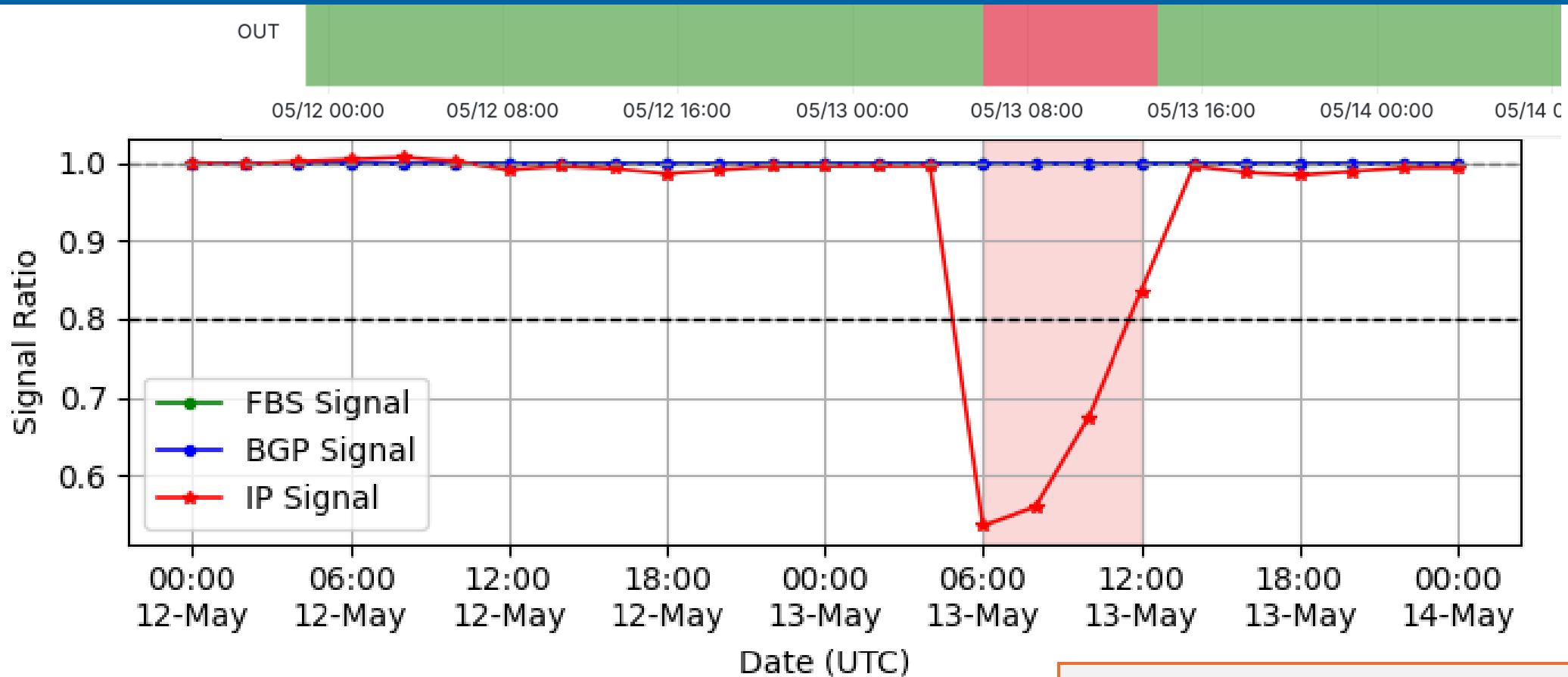
13.05.2022

Вторгнення окупантів в серверну компанії "СТАТУС".

Breaking of the russian occupants into the "STATUS" company's room.



#8 Dataset provides insights into smaller regional providers in Kherson.

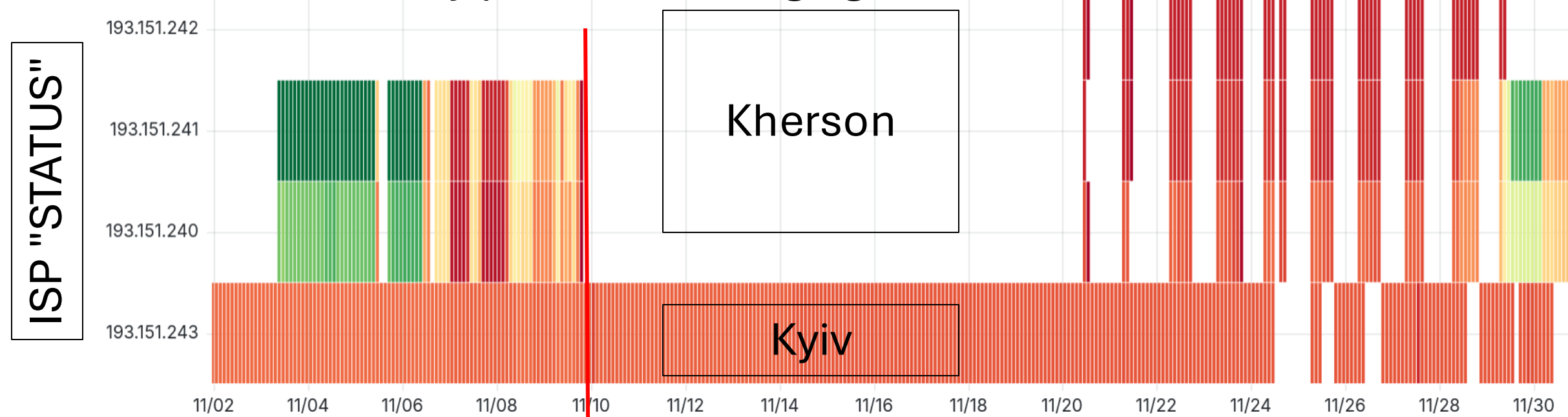


CountryMonitor: Decline in responsive IPs for ASN 25482 (STATUS ISP) at the same time as the incident

Incident: Russian soldiers search Status ISP and take infrastructure offline

#8 Dataset provides insights into smaller regional providers in Kherson.

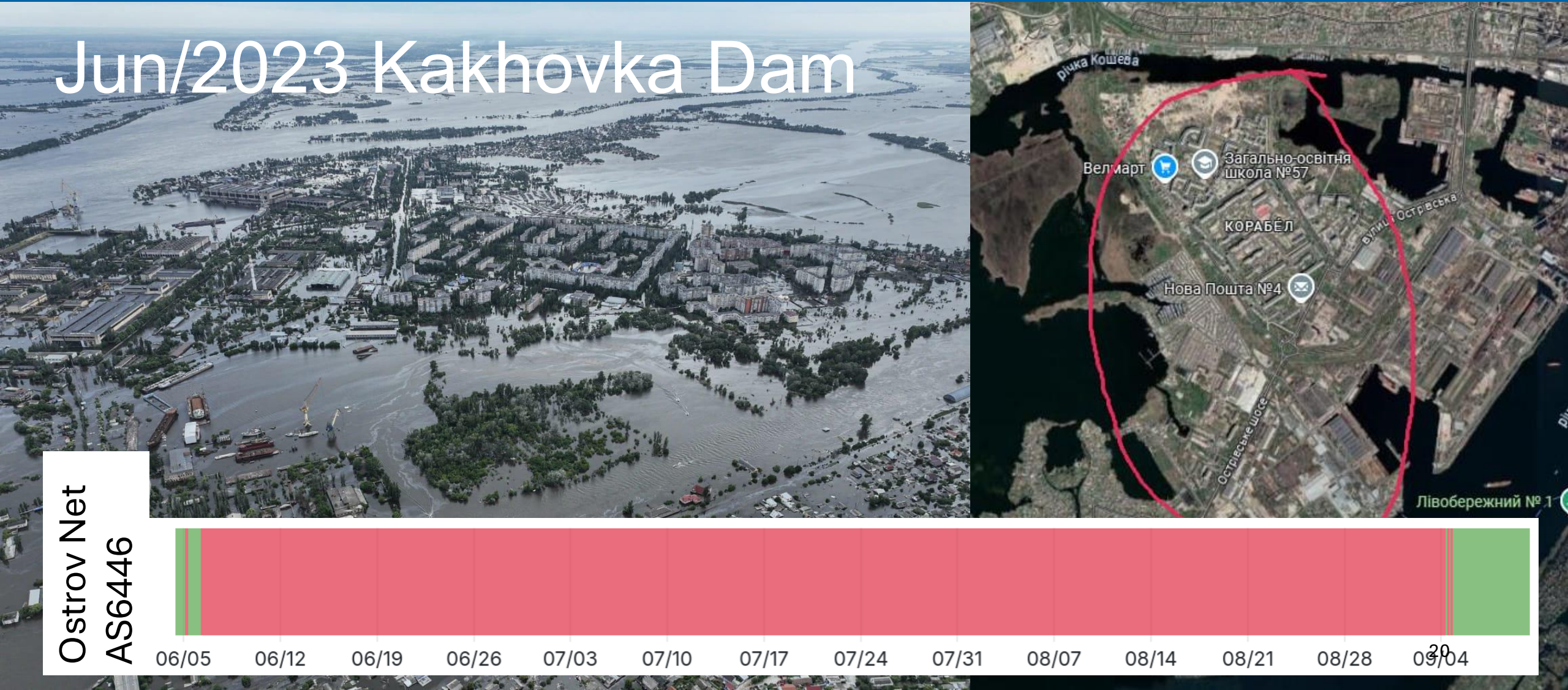
Nov 2022: Kherson city powered through generators



Russian Troops destroy critical infrastructure (TV Tower, Heat/Power Plant during retreat)

#8 Dataset provides insights into smaller regional providers in Kherson.

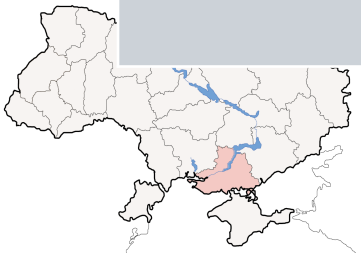
Jun/2023 Kakhovka Dam



CountryMonitor - Takeaways



	Singla et al.	Klick et al.	IODA	CountryMonitor
Protocols	Modbus, DNP3	60+	ICMP	ICMP
Signal	IP based	IP based	Block based (Sampling)	Block based (no Sampling)
Interval	24 Hours	4 Hours	10 Minutes	2 Hours
War Coverage	6 Months 2022	Until March 2023	Since 2022	Since 2022
Outage Location	IP Geolocation	Static IPs	IP Geolocation	Regional Classification



Short-lived outages $\geq 2h$
~~Static IPs only?~~ ✓

Sampling? Small Providers ✓
Geolocation Precision ✓

Acknowledgements:

Alexander Kovalenko – Status ISP

Dipl.-Ing. Florian Holzbauer

Researcher & PhD Candidate

University of Vienna

florian.holzbauer@univie.ac.at



countrymonitor.github.io
Research Access -> Oct 28th 2025

#BONUS Dataset provides insights into smaller regional providers in Kherson.

Damage to Mykolaiv cable

> multi day outage in Kherson (May 1 to May 4)

Rerouting over Russian

upstreams during summer

> Short outages for regional operators; then routed over Miranda

> Non-regionals often disconnected

Kakhovka

dam impacts ISPs in flooded areas

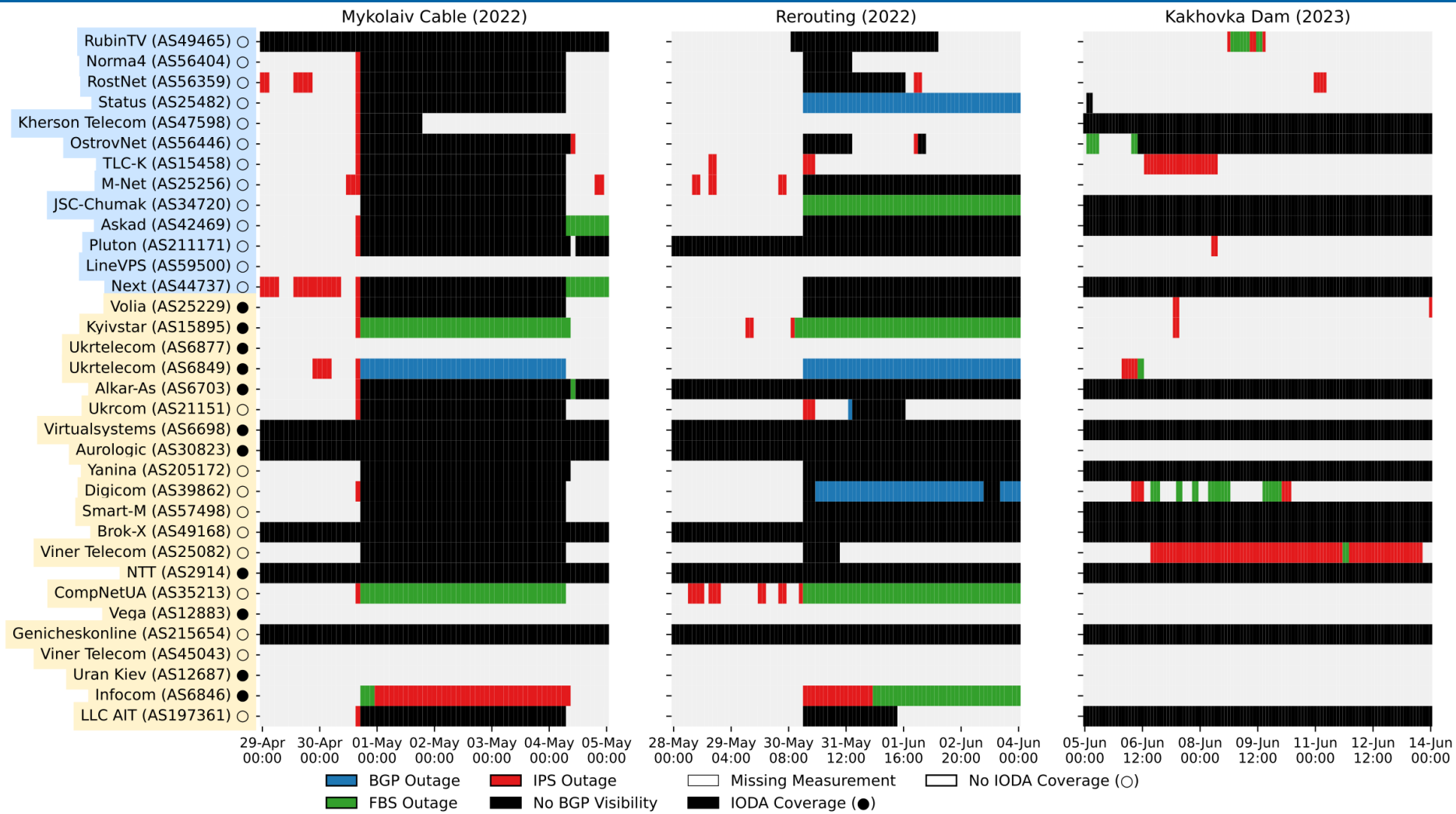


Figure 9: Three events with significant impact on ISPs in frontline region Kherson.

#BONUS Partial outages become visible when extending Full Block Scans with an IP signal.

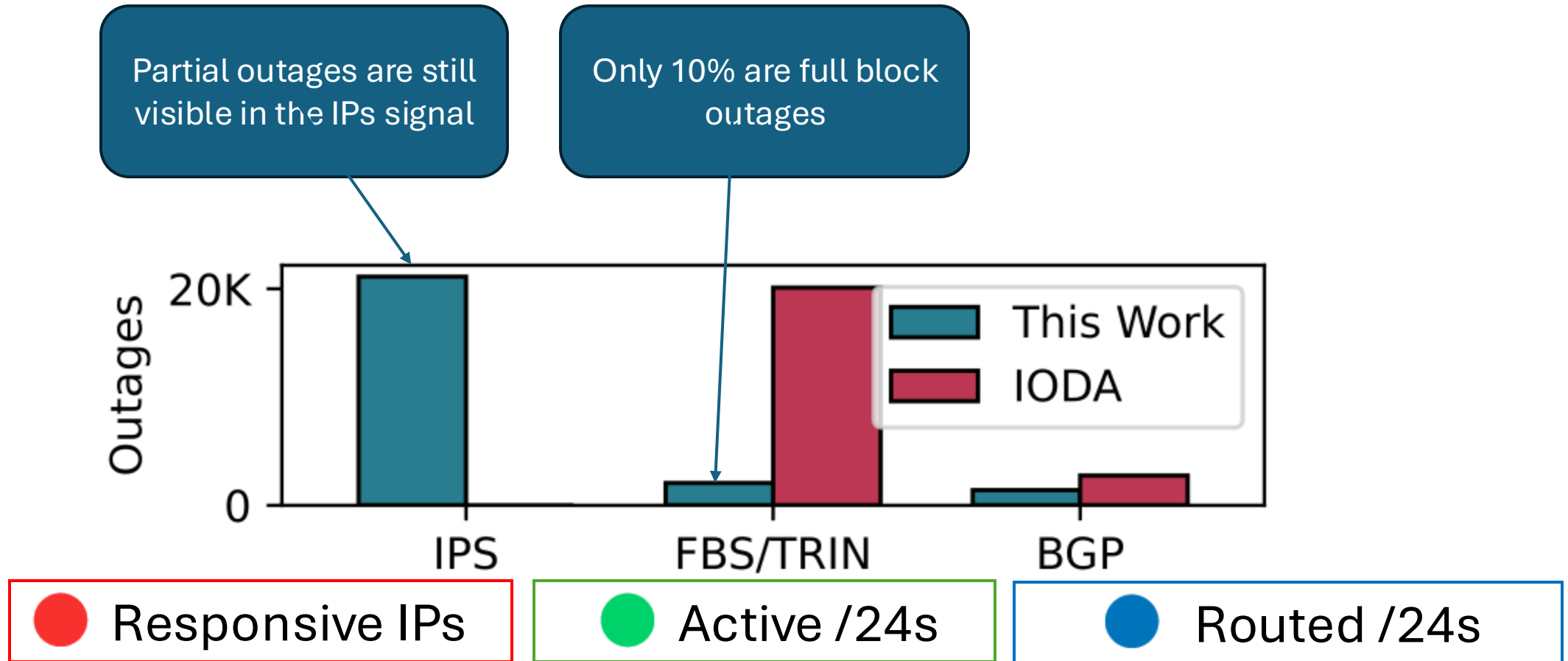


Figure 10: Outages per signal for a set of 182 ASes for which both report outages & cover > 0.9 of routed IPs

Detecting Internet Disruptions

Quan - Trinocular (2013)

IPv4 wide; /24 based
up to 15 probes every 11
minutes
ever active IPs

Baltra – Full Block Scans (2019)

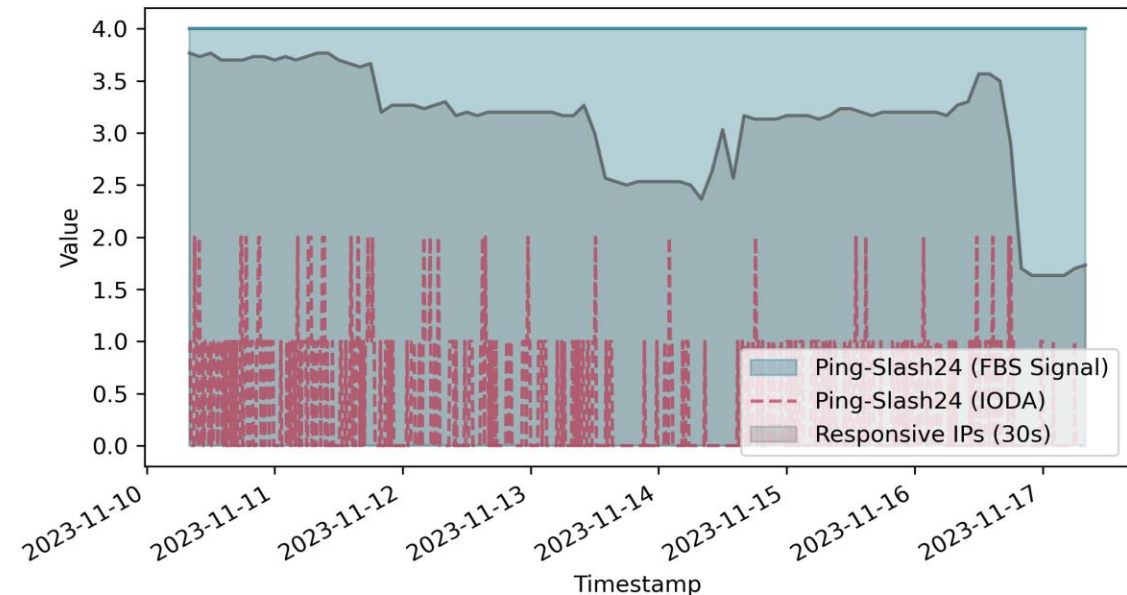
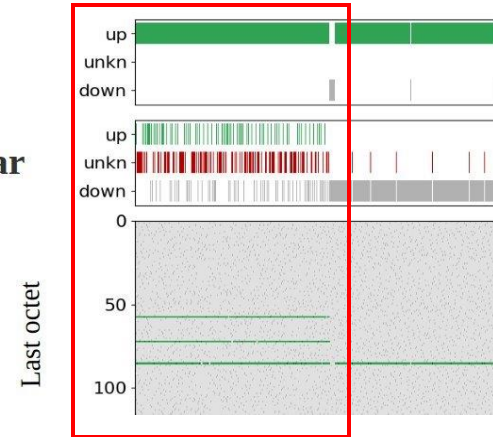
aggregates multiple Trinocular rounds;
Avoid many down events in sparse /24s
Filter maintenance activity through *ISP Availability Sensing (/24 down; IPs still up in another /24)*

Richter (2018)

Disruption != Outage
Trinocular includes many disruptions
Solution: Filter /24s with more than 5
outages in 3 months reduces number of
false outages

FBS

Trinocular



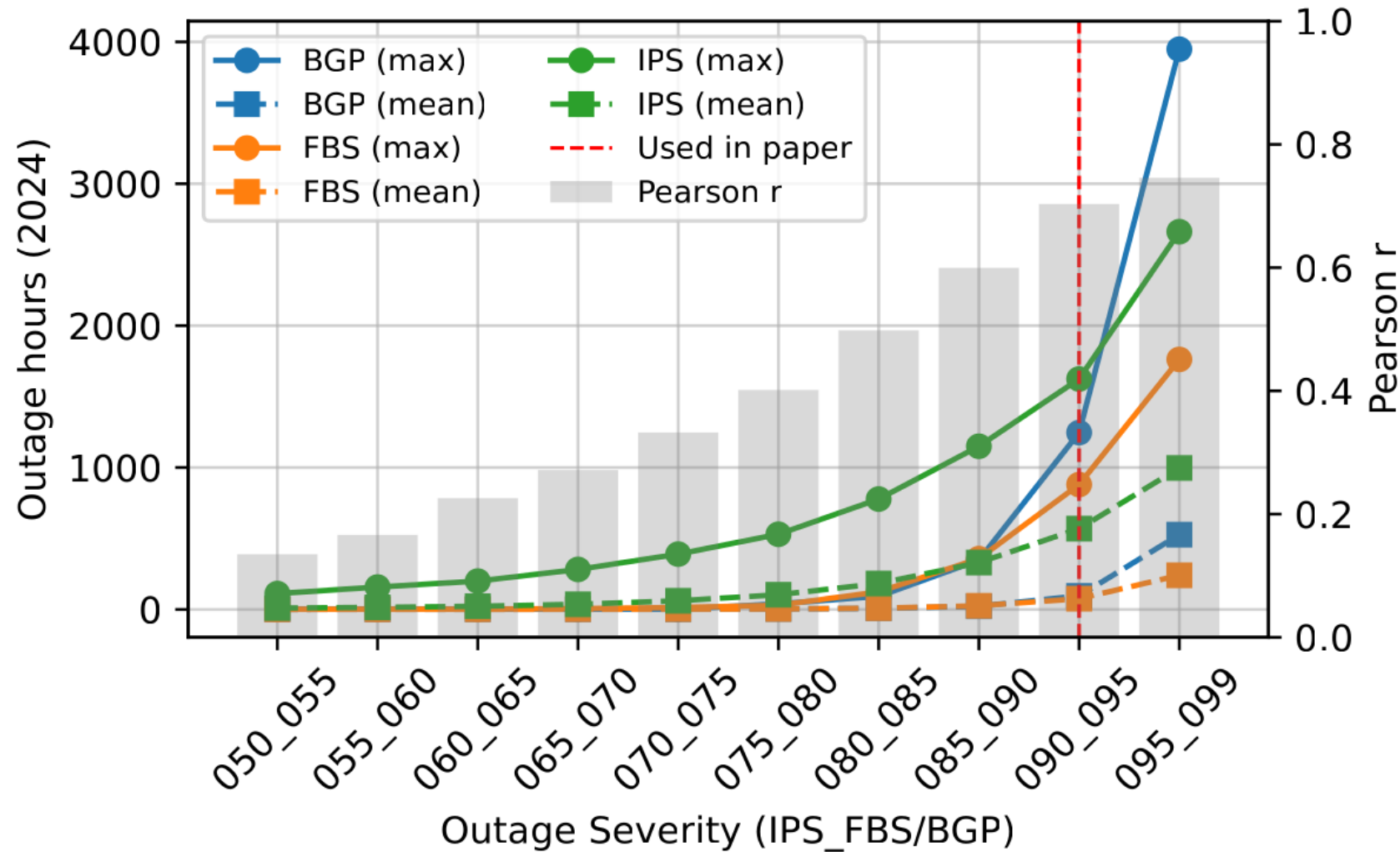
Parameters & Threshold Sensitivity/Severity

Level	BGP ★	FBS ■	IPS ▲
AS	< 95%	< 80% (if IPS < 95%)	< 80%
Regional	< 95%	< 95% (if IPS < 95%)	< 90%

Solution: Baltra – ISP Availability Sensing (2019)

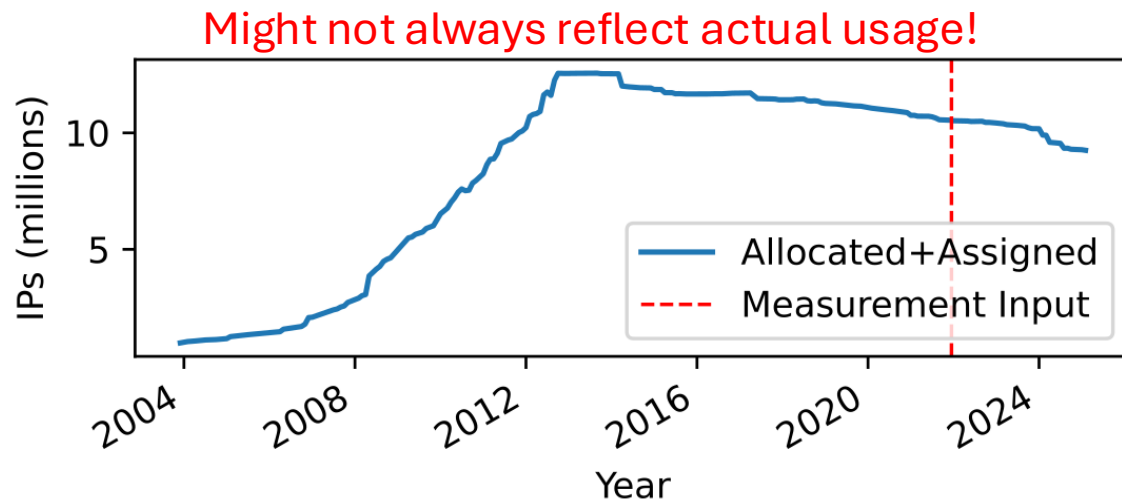
Filter maintenance activity > additionally responsive IPs have to go down by 5% or more

Parameters & Threshold Sensitivity/Severity



What is the size of the Ukrainian IPv4 address space?

RIPE Allocations



- 3,085 IPv4 ranges (~10.5M IPs) in 2022; trend declining
 - 98% still exist in 2025
 - 87% allocated to Ukraine
 - 13% changed country codes
 - one-third to Russia
 - Still valid measurement targets

IP Geolocation

- Leased Prefixes?
 - 3.3% of delegated IPs located outside the country
 - Included in our measurements ✓
 - Not valuable measurement targets?
 - Reverse case: 7.4% of IPs locate to Ukraine but delegated to other countries
 - Not included in our measurements ✗